$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \widetilde{D}\left[\hat{\sigma}_z, [\hat{\sigma}_z, \hat{\rho}_{\mathcal{S}}(t)]\right], \tag{5.149}$$